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Department of Computer Science

Title: Image Enhancement System

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Course: Digital Image Processing

Lab: 6

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1. Objective:

The objective of this project is to develop an integrated Digital Image Processing system capable of performing fundamental operations, including image acquisition, spatial/intensity resolution adjustment, geometric transformations, and contrast enhancement using Histogram Equalization and Power-Law transformations.

2. Methodology:

Phase 1: Image Acquisition & Pre-processing

The system begins by loading raw image data (RGB). To reduce computational complexity and focus on intensity analysis, images are converted to a 2D grayscale matrix.

- **Key Tool:** cv2.cvtColor

Phase 2: Sampling and Quantization

This phase analyzes how digital constraints affect image quality.

- **Sampling:** Reducing the pixel count (spatial resolution).
- **Quantization:** Reducing the bit-depth (intensity resolution).
- **Observation:** Lowering bits leads to "False Contouring."

Phase 3: Geometric Transformations

We manipulate the spatial coordinates of pixels. The system performs Rotation, Translation, and Shearing. We then apply **Inverse Transformations** to attempt to restore the original image, observing data loss at the boundaries.

Phase 4: Intensity Transformations

We apply point processing techniques:

- **Image Negative:** Inverts intensities.
- **Log Transform:** Expands dark pixel values.
- **Gamma Correction:** Adjusts brightness non-linearly to reveal hidden details.

Phase 5: Histogram Processing

The system calculates the global contrast of an image. **Histogram Equalization** is used to redistribute pixel intensities, effectively "stretching" the contrast of washed-out images.

Phase 6: Final Integrated System

A multi-stage pipeline is established: **Grayscale → Histogram Equalization → Gamma Tuning.** This represents the optimized flow for automated image enhancement.

3. Code Snippets:

Lab 01: Image Acquisition & Pre-processing

Focus: Converting the 3D RGB array into a 2D Intensity matrix.

Code:

```
# Convert BGR (OpenCV default) to Grayscale
img_gray = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2GRAY)
# Accessing the underlying matrix (First 5x5 pixels)
print(img_gray[:5, :5])
```

Lab 02: Sampling & Quantization

Focus: Spatial reduction and bit-depth mapping.

Code:

```
# Sampling: Reducing spatial resolution by 50%
low_res = cv2.resize(img_gray, None, fx=0.5, fy=0.5)
# Quantization: Mapping 256 levels to 4 levels (2-bit)
level = 4
quantized = (np.floor(img_gray / (256/level)) * (256/level)).astype(np.uint8)
```

Lab 03: Geometric Transformations

Focus: Affine matrices for spatial mapping.

Code:

```
# Rotation: Creating a 2D rotation matrix for 45 degrees
M = cv2.getRotationMatrix2D((cols/2, rows/2), 45, 1)
rotated = cv2.warpAffine(img_gray, M, (cols, rows))
# Shearing: Applying a horizontal tilt
M_shear = np.float32([[1, 0.2, 0], [0, 1, 0]])
sheared = cv2.warpAffine(img_gray, M_shear, (int(cols*1.2), rows))
```

Lab 04: Intensity Transformations

Focus: Non-linear mapping using Power-Law (Gamma).

Code:

```
# Image Negative: s = (L-1) - r
negative = 255 - img_gray
# Gamma Correction: s = c * r^gamma (using a Look-Up Table)
invGamma = 1.0 / 1.5
table = np.array([((i / 255.0) ** invGamma) * 255 for i in np.arange(0, 256)]).astype("uint8")
gamma_output = cv2.LUT(img_gray, table)
```

Lab 05: Histogram Processing

Focus: Global contrast enhancement via equalization.

Code:

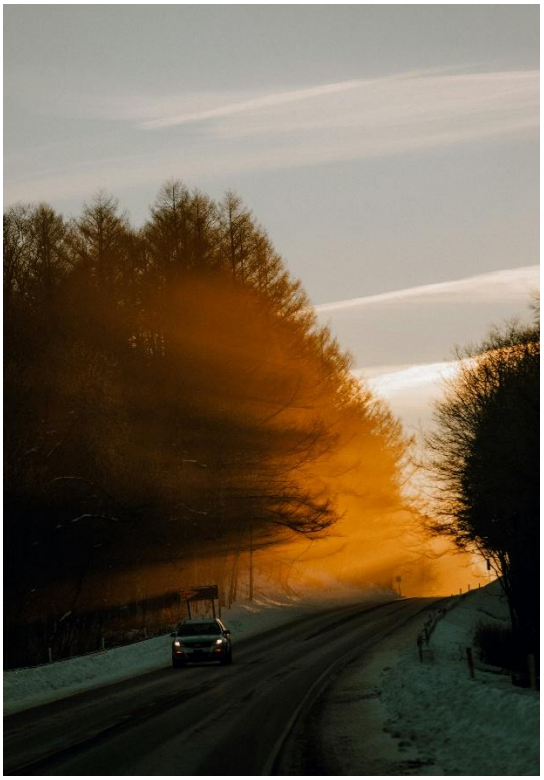
```
# Histogram Equalization: Spreading the intensity distribution
img_equ = cv2.equalizeHist(img_gray)
# Visualization: Flattening the 2D array for plotting
plt.hist(img_gray.ravel(), bins=256, range=[0, 256])
```

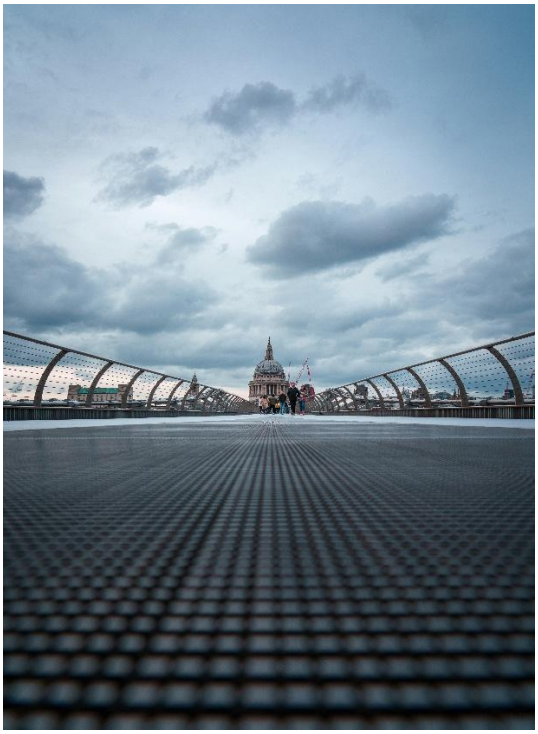
Lab 06: Final Integrated System

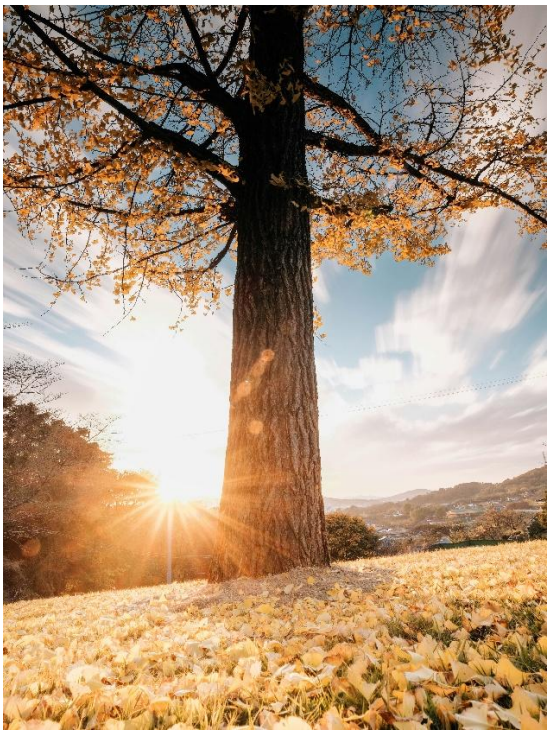
Focus: The logical pipeline of operations.

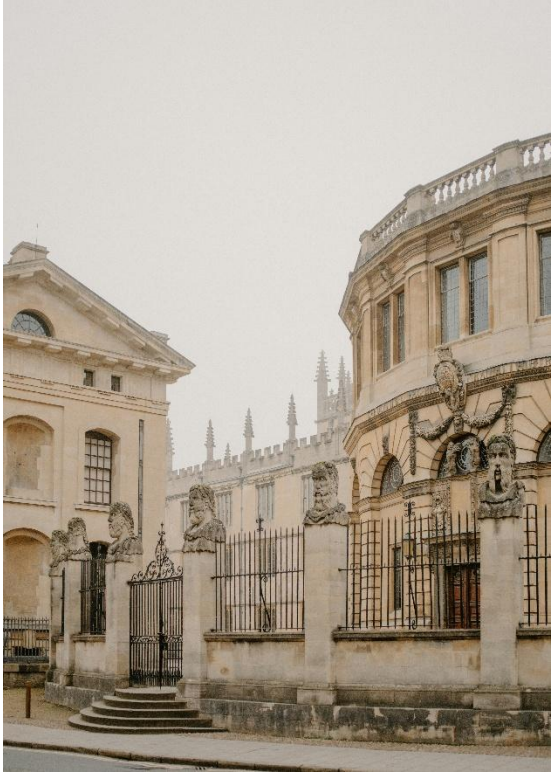
```
# The Enhancement Pipeline
def integrated_system(img_path):
    gray = cv2.cvtColor(cv2.imread(img_path), cv2.COLOR_BGR2GRAY)
    equ = cv2.equalizeHist(gray) # Step 1: Global Contrast
    final = cv2.LUT(equ, gamma_table) # Step 2: Fine-tuning
    return final
```

4. Input images:

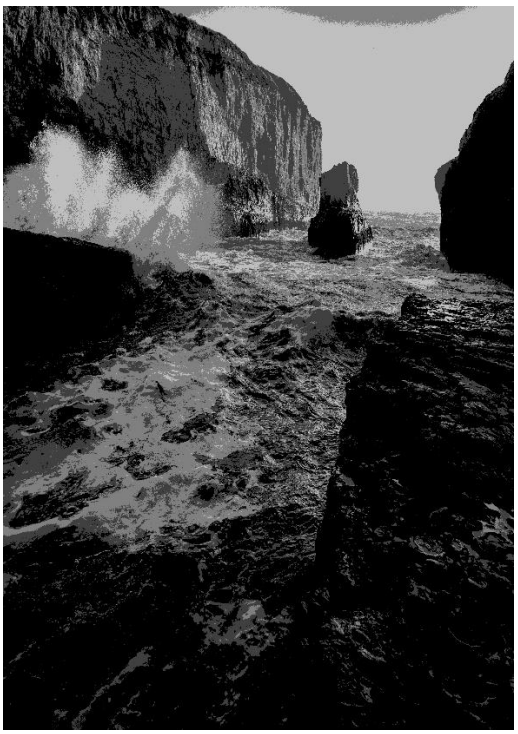


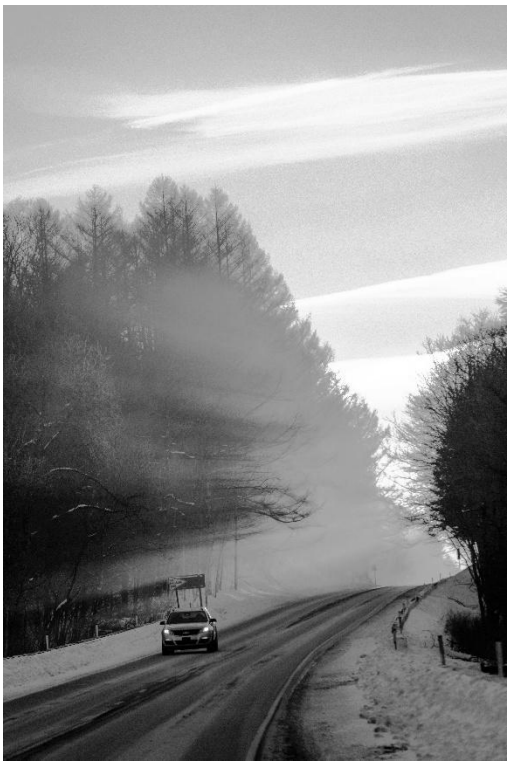






5. Output Images:





6. Observations & Analysis

Based on the experiments conducted across the six labs, the following technical observations were made:

6.1. Spatial vs. Intensity Resolution

- **Observation:** When reducing the sampling rate (Scale 0.25x), the image became "pixelated," losing fine details like sharp edges and textures.
- **Analysis:** This confirms that **Spatial Resolution** is directly tied to pixel density. Furthermore, in **Quantization**, reducing the bit-depth to 2-bit (4 levels) caused "False Contouring"—visible bands of gray instead of smooth gradients. This proves that at least 8-bit depth is required for natural human perception of monochrome images.

6.2. Geometric Information Loss

- **Observation:** Rotating an image and then performing an "Inverse Rotation" did not perfectly restore the original pixels at the corners.
- **Analysis:** This occurs because pixels that move outside the image boundary during the first transformation are "clipped" and replaced by black pixels (value 0). This data is permanently lost and cannot be recovered by simple inverse math.

6.3. Enhancement Efficacy

- **Observation:** While **Log Transformations** (Lab 04) effectively brightened very dark regions, they often reduced the overall contrast of the image.
- **Analysis:** In contrast, **Histogram Equalization** (Lab 05) provided a more "balanced" enhancement by spreading the intensity distribution. However, when combined with **Gamma Correction** in Lab 06, the results were superior as it allowed for fine-tuning the brightness after the contrast was normalized.

7. Final Conclusion

The Digital Image Processing (DIP) system developed through this project successfully demonstrates a complete pipeline for image analysis and enhancement.

Key takeaways from the project include:

1. **Preprocessing is Essential:** Converting to Grayscale is a necessary step for mathematical simplicity in intensity-based operations.
2. **Trade-offs in Resolution:** Digital images are limited by their sampling and quantization rates; balancing file size and visual fidelity is a core challenge in DIP.
3. **The Hybrid Advantage:** No single transformation (Log, Gamma, or Equalization) is a "silver bullet." The **Integrated System (Phase 6)** proved that a multi-stage approach—normalizing contrast globally and then fine-tuning brightness locally—produces the most professional and visually accurate results.

This project provides a foundational framework that can be extended into more advanced domains such as **Medical Imaging**, **Satellite Data Analysis**, or **Computer Vision**.